



Growth, Feed Utilization and Water Productivity of *Heteropneustes fossilis* Under Different RAS Stocking Densities

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ABSTRACT

A 75-day experiment was conducted to evaluate the growth and physio-metabolic performance of stinging catfish in a recirculating aquaculture system (RAS). The system was modified with the aim of cost-effectiveness, where a significant portion of electricity costs was reduced by maintaining gravitational water flow, and a sedimentation tank was installed to reduce the filtration load on the filters. The system was designed and installed at the Central Institute of Fisheries Education, Motipur, where fish were stocked in nine covered polythene tanks, each with a capacity of 30,000 L. During the experiment, water flow was maintained at 11.7 L s^{-1} , and water was exchanged at a rate of 10% at 15-day intervals. Fish weighing $4.37 \pm 0.27 \text{ g}$ were stocked at different densities, i.e., 200 fish m^{-3} (T200), 250 fish m^{-3} (T250), and 300 fish m^{-3} (T300), under optimal water quality conditions. The results indicated the highest growth performance in T200 and T250, and a similar trend was observed in feed conversion ratio, feed efficiency ratio, protein efficiency ratio, and survival rate. Economic and water-use efficiency analyses demonstrated that T250 was sustainable and economically feasible, although the values were statistically similar across treatments. Therefore, it can be concluded that sustainable culture of Singhi catfish can be achieved at a stocking density of 200-250 fish m^{-3} in an innovative and commercial RAS.

1. Introduction

Supply of quality nutrition to the burgeoning human population is a major global challenge, particularly in the context of limited land resources. In the current scenario, aquaculture emerges as a promising solution for securing food and meeting nutritional demands [1-2]. The aquaculture sector has rapidly emerged as the fastest-growing food-producing sector; however, the decline in per capita land availability is driving farmers toward the intensification of aquaculture production systems [3]. Intensified aquaculture systems enhance productivity and support ease of management, which highly motivates farmers to adopt sustainable, high-yielding technologies such as recirculating aquaculture systems (RAS) and biofloc technology [4-5]. The implementation of RAS has integrated modern aquaculture approaches through several means, including intensive stocking, biosecurity, and water filtration, thereby reducing water usage in a closed-loop system [6-7]. These systems rely on water filtration, and improper filtration methods may lead to the accumulation of harmful waste compounds such as ammonia and nitrite. Modern RAS should focus on dynamic system modeling, which offers a more integrative approach by examining the biological, chemical, and physical components of the system. These systems are particularly effective in facilitating adaptive water management and optimizing system design in aquaculture [6]. RAS supports the sustainable production of commercially important species such as Barramundi, Cobia, Silver Pompano, Tilapia, Pearl Spot, Pangasius, and Rainbow Trout. Recent studies also highlight the economic feasibility of culturing catfishes such as Ompok pabda and *Heteropneustes fossilis* in RAS [8-9]. *Heteropneustes fossilis* (Singhi) is a high-value fish in India due to its nutritional benefits and palatability, and its fast growth along with higher stress tolerance makes it suitable for intensive aquaculture systems [10-11]. Culturing Singhi in RAS promotes growth, reduces operational costs, and limits water use [12]. It also contributes in efficient use of resources, responsible aquaculture management, and year-round production, making it a lucrative option in the fish production sector [13].

In intensive aquaculture practices, stocking density is a critical factor that significantly influences the growth and welfare of fish [14]. High stocking density can induce physiological stress responses, immunosuppression, growth retardation, reduced survival, and increased susceptibility to diseases [15]. Stocking density is solely species-dependent and should be determined according to the tolerance of the species and the type of culture system employed, which typically ranges between 10 and 100 kg m^{-3} [16-18].

The present study was designed to optimize stocking density for the commercial culture of *H. fossilis* in RAS. Through this experiment, we evaluated the effects of different stocking densities on feed utilization efficiency, growth performance, survival, and economic returns of the commercially important high-value catfish *H. fossilis*.

2. Materials and Methods

2.1 Experimental location, design, and setup

A 75-day experiment was conducted at ICAR-CIFE, Motipur, Bihar, to evaluate the ideal stocking density for the culture of Singhi in an innovative intensified aquaculture system, i.e., RAS. The experiment was carried out in nine closed circular RAS tanks with a capacity of 30,000 L each. Water level was maintained at 28,000 L. A completely randomized design was followed, comprising three treatments: T200 (200 fish m^{-3}), T250 (250 fish m^{-3}), and T300 (300 fish m^{-3}), each stocked in triplicate. The filtration unit was set up with a drum filter as a mechanical filter, and five chambers were set up as a biological filtration unit with different bio-media, i.e., gravel, sand, bio-ball, and bio-ring, and continuous aeration was provided. The bio-media were installed in tanks 60 days prior stocking of fishes for better functioning. A daily flow rate was adjusted at 11.7 L s^{-1} to recycle total water through the filtration unit four times per day.

2.2 Specification of RAS Model

The RAS system is based on the core principle of adapting the system toward an energy-efficient and cost-effective sustainable approach for semi-intensive and intensive aquaculture. Excess and heavy suspended solids, along with feed and fecal matter, are removed just before filtration through a sink tank. This reduces the load on the drum filter, and under lower loading conditions, the drum filter may consume less electricity, thereby reducing operational costs. The removal of sludge also decreases the load on both mechanical and biological filtration units, significantly eliminating the frequent washing and management needs. Gravity flow is assigned to the system for refilling the filtered water.

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2.3 Experimental animal and diets

Heteropneustes fossilis juveniles (2-6 g) were procured from a local vendor (a wholesaler dealing in supply of wild-caught fishes from the floodplains of Bihar) and transported to ICAR-CIFE, Motipur, in oxygenated tanks. They were acclimated in 30,000 L capacity tanks for two weeks and fed a commercial diet containing 40% crude protein and 8% lipid (0.8 mm pellet diameter; Godrej Agrovet, India) at 2% of body weight, four times daily. After two weeks, fish with a body weight of 4-5 g were distributed into RAS tanks across three treatment groups in triplicate, at stocking densities of 200 fish m⁻³, 250 fish m⁻³, and 300 fish m⁻³, respectively [19]. After stocking, the fish were fed a commercial diet of 32% crude protein and 5% lipid (1 mm pellet diameter; Godrej Agrovet, India) at 4% of body weight, twice daily.

2.4 Physicochemical water quality parameters

The physicochemical parameters of water in all experimental tanks were observed weekly using standard instruments and spectrophotometric analytical methods. Water temperature was measured using a mercury thermometer (Aggarwal Crockery & Scientific Stores). Water quality parameters such as pH, dissolved oxygen (mg L⁻¹), total ammonia nitrogen (mg L⁻¹), nitrite (mg L⁻¹), and nitrate (mg L⁻¹), were assessed weekly using standard spectrophotometric protocols [20].

2.5 Sampling

Following the completion of 75 days, 15 fish were randomly selected from each tank and anesthetized using a clove oil immersion method at a concentration of 50 µL L⁻¹ to evaluate growth and whole-body composition.

2.6 Growth performance, feed utilization parameters, and survivability

Fish growth was monitored at 15-day intervals and evaluated using various growth parameters such as final body weight, weight gain, percentage weight gain, specific growth rate, and average daily growth. Additionally, feed utilization parameters such as feed conversion ratio, feed efficiency ratio, and protein efficiency ratio were calculated for all treatments. Survival rate was also determined.

- 1) Weight gain (g) = Final fish body weight (g) / Initial fish body weight (g)
- 2) Weight gain (%) = [Final fish body weight (g) - Initial fish body weight (g) / Initial fish weight (g)] × 100
- 3) Specific growth rate (% day⁻¹) = [Ln final fish body weight (g) - Ln initial fish body weight (g) / Experimental period (days)] × 100
- 4) Average daily growth (% day⁻¹) = [Final fish body weight (g) - Initial fish body weight (g) / Experimental period (days)] × 100
- 5) Feed conversion ratio = Feed intake (dry weight in g) / Weight gain (g)
- 6) Feed efficiency ratio = [Weight gain (g) / Feed intake (g)]
- 7) Protein efficiency ratio = [Weight gain (g) / Protein intake (g)]
- 8) Survival rate (%) = [Total number of live fish harvested / Total number of fish stocked] × 100

2.7 Proximate composition analysis of fish

The whole-body proximate composition of the experimental fish was determined according to standard procedures [21] on a dry matter basis. Moisture content was analysed by drying the samples at 105°C in a hot-air oven (Prayag Scientific, India). Crude protein content was determined by the micro-Kjeldahl method using an automatic protein analyser (Kjel COM 6M AT, Tulin, India). Total lipid content was estimated using the Soxhlet extraction method with an automatic Soxhlet apparatus (Sox-6, Tulin, India). Total ash content was determined using a muffle furnace (PLF 140, Labindia, India) at 550°C for 6 h. The following standard formula were used to calculate moisture, crude protein, total lipid, and total ash:

- 1) Moisture (%) = [Sample weight before drying (g) - Sample weight after drying (g) / Sample weight before drying (g)] × 100
- 2) Crude protein (%) = [Total nitrogen content (%) × 6.25]
- 3) Ether extract (%) = [Weight of the dried receiver flask plus extracted lipid (g) - Weight of the dried receiver flask (g) / Weight of the dried sample (g)] × 100
- 4) Total ash (%) = [Ash weight (g) / Dry sample weight (g)] × 100

2.8 Economics analysis

The economic indices were calculated at the end of the experiment using conventional methods as mentioned by [9,22-28].

- 1) Cost of seed = [Quantity of seed (numbers) stocked in a tank × cost per seed]
- 2) Cost of feed = [Quantity of feed (kg) used × cost per kg feed]
- 3) Cost of manpower = [(Number of employees × Culture duration × Salary per day) / Number of tanks]
- 4) Electricity cost = [Cost of electricity (kW-hour) × Total kW used during culture / Number of tanks]
- 5) Annual economic depreciation for infrastructure and machineries = [(Costs of tanks, infrastructure, instruments) / Approximate economic life]
- 6) Land rent (1 decimal = 40.47 m²) = [Farm area (in decimal) × Rent of the land per month (at 1.65 decimal) × Culture duration / Number of tanks]
- 7) Total investment or cost of production = [Operating cost (A) + Capital expenditure (B)]
- 8) Net income or profit = [Gross income - Total investment]
- 9) Return on investment % = [(Net profit / Total production cost) × 100]
- 10) Annual net profit = [Net income × Number of culture cycle in a year]

2.9 Water productivity

The water productivity indices were calculated using conventional and established methods as reported by [27,29-32].

- 1) Total water use (TWU) = [Initial water volume in m³ + Regulated water flow to compensate evaporation and bottom cleaning in m³]
- 2) Consumptive water use (CWU) = [TWU - Final water volume during harvest in m³]
- 3) Consumptive water use index (CWUI) = [CWU / number of fish survived]
- 4) Total water use index (TWUI) = [TWU / number of fish survived]
- 5) Physical water productivity (PWP) = [Total yield of fish in kg / Total volume of water in m³]
- 6) Gross total water productivity (GTWP) = [Total economic value of the product in USD / Total volume of water used in m³]
- 7) Net total water productivity (NTWP) = [(Total Economic value of the product in USD - Production cost in USD) / Total volume of water used in m³]

2.10 Statistical analysis

The data analysis was done in Statistical Package for the Social Sciences (SPSS), version 26.0, for Windows, by applying one-way analysis of variance (ANOVA) and expressed in mean ± standard error. Tukey's multiple range test was used for the post-hoc analysis.

3. Results and Discussion

3.1 Growth performance

The growth performance of Singhi during the 75-day experiment is assessed and presented in Table 1. Significantly higher ($P \leq 0.05$) growth performance was observed in the T200 group, with a final body weight of 20.79 ± 0.32 g, weight gain percentage of $379.35 \pm 26.72\%$, and average daily growth of 0.219 ± 0.01 g. A significant reduction in growth was observed with increasing stocking density. However, survivability was statistically similar ($P > 0.05$) among all experimental groups.

3.2 Feed utilization and survivability

The experiment showed significantly higher ($P \leq 0.05$) feed utilization parameters, i.e., feed conversion ratio (0.80 ± 0.06), protein efficiency ratio (3.94 ± 0.25), and feed efficiency ratio (1.26 ± 0.09), in the T200 group, as mentioned in Table 1.

Table 1 Growth, feed utilization, and survival of *Heteropneustes fossilis* at different stocking densities after a 75-day culture period in a recirculating aquaculture system.

Parameters/Treatments	T200	T250	T300	P value
Initial body weight (g)	4.37±0.27	4.40±0.15	4.33±0.09	0.969
Final body weight (g)	20.79±0.32 ^c	18.97±0.38 ^b	16.38±0.43 ^a	0.001
Weight gain (g)	16.42±0.34 ^c	14.57±0.35 ^b	12.04±0.43 ^a	0.001
Weight gain (%)	379.35±26.72 ^b	331.85±14.17 ^{ab}	278.17±11.69 ^a	0.025
Specific growth rate (g)	2.09±0.08 ^b	1.95±0.04 ^{ab}	1.77±0.04 ^a	0.021
Average daily growth (g)	0.219±0.01 ^c	0.194±0.01 ^b	0.161±0.01 ^a	0.001
Feed conversion ratio	0.80±0.06 ^a	0.91±0.04 ^{ab}	1.08±0.04 ^b	0.017
Protein efficiency ratio	3.94±0.25 ^b	3.41±0.15 ^{ab}	2.87±0.12 ^a	0.016
Feed efficiency ratio	1.26±0.09 ^b	1.11±0.05 ^{ab}	0.93±0.04 ^a	0.025
Survival rate (%)	87.13±0.64	84.10±1.71	82.58±2.14	0.212

Data are expressed as mean $\pm$ standard error, $n = 3$ (one-way analysis of variance); mean values in the same row with different superscripts differ significantly ($P < 0.05$). T200: 200 fish m^{-3} , T250: 250 fish m^{-3} , T300: 300 fish m^{-3} .

In the current scenario, aquaculture is increasingly focused on intensification due to rising demand in national and global markets [33]. Intensification of aquaculture using hardy and climate-resilient species can provide higher returns to farmers [34]. RAS is one of the important intensive aquaculture systems widely adopted by farmers in India due to its high per capita productivity and suitability for a wide range of commercially important fish species [5]. Singhi, being a commercially important, climate-resilient, and hardy species, is highly suitable for RAS-based systems, but its stocking density should be optimized for better economic returns [35]. Stocking density is one of the important factors optimizing the growth and survivability of fish [36].

To understand the effect of stocking density in RAS-based systems, Singhi were stocked at three different densities, i.e., 200 fish m^{-3} , 250 fish m^{-3} , and 300 fish m^{-3} . Growth performance showed a decreasing trend with increasing stocking density, with the highest weight gain percentage in T200 and the lowest in T300. Similarly, feed conversion ratio and average daily growth values were highest in the T200 group. This output may be associated with crowding stress and the associated physiological changes in the experimental animals [37-38]. Similar growth retardation associated with higher stocking density has been reported in tambaqui (*Colossoma macropomum*), ayu (*Plecoglossus altivelis*), and tiger puffer (*Takifugu rubripes*) [39-41]. However, increase in stocking density did not affect the survivability of the fish, which signifies suitability of the selected species in the system as well as better water quality management of the system [42].

In intensified aquaculture systems, feed plays a crucial role in meeting the nutritional requirements of fish. Balanced nutrition may provide optimal growth and helps achieve desirable output within a specific culture period. In this experiment, fish were fed a commercial diet containing 40% crude protein and 10% lipid, which is considered ideal for grow-out culture [43]. Improved feed conversion ratio, protein efficiency ratio, and feed efficiency ratio were observed at lower stocking density (T200), which shows a decreasing trend with increased stocking density. Increase in stocking density may lead to competition among fish for feed, which leads to impaired digestion, thereby reducing growth [14,44].

3.3 Whole-body proximate composition

The whole-body proximate composition, including moisture, crude protein, ether extract, and ash content of the experimental fish cultured in RAS at different stocking densities, did not differ significantly among the treatments, as presented in Table 2.

Table 2 The whole-body composition (% dry matter basis) of *Heteropneustes fossilis* after a 75-day culture period in a recirculating aquaculture system.

Treatments	Moisture	Crude Protein	Ether Extract	Ash
T200	73.95±0.51	63.70±0.21	13.90±0.27	9.76±0.28
T250	73.83±0.22	64.17±0.55	13.87±0.19	9.77±0.38
T300	74.11±0.29	64.13±0.35	14.17±0.20	9.90±0.31
P value	0.865	0.669	0.605	0.944

Data are expressed as mean $\pm$ standard error, $n = 3$ (one-way analysis of variance); mean values did not show significant differences ($P > 0.05$). T200: 200 fish m^{-3} , T250: 250 fish m^{-3} , T300: 300 fish m^{-3} .

The whole-body composition of Singhi catfish cultured in RAS did not show any significant variation in protein, lipid, and ash content, which signifies that the fish were well nourished even at higher stocking densities. Higher stocking density may impact the nutrient composition of fish, which may lead to reduced whole-body lipid content and a decrease in growth due to crowding stress [45]. The protein content is sometimes stocking density-dependent and may either increase or remain unchanged depending on the species and diet [46-47]. Conversely, lower stocking densities are associated with higher lipid accumulation in muscle and improved growth performance [48]. High stocking density can also negatively affect nutrient composition by increasing stress, which may disrupt lipid metabolism and lead to reduced fat content in the fish [49]. The statistically similar proximate composition across all treatments may be attributed to the better culture environment associated with better water circulation and uniform water quality throughout the system.

3.4 Water quality and water circulation parameters

Physicochemical parameters of water, such as temperature, pH, total ammonia nitrogen ($mg L^{-1}$), nitrite-N ($mg L^{-1}$), and nitrate-N ($mg L^{-1}$), were estimated weekly at 9 AM. During the experimental period, water temperature, pH, total ammonia, nitrite, and nitrate were maintained in the range of $28^{\circ}C$ - $33^{\circ}C$, 7.6-8.4, 0-0.05 $mg L^{-1}$, 0-0.01 $mg L^{-1}$, and 0-0.01 $mg L^{-1}$, respectively, as mentioned in Table 3. The water quality in all experimental tanks remained within optimal range throughout the experiment.

Table 3 Physicochemical parameters of water used to rear experimental fish for 75 days in a recirculating aquaculture system.

Parameters	T200	T250	T300
pH	7.6-8	7.6-8.4	7.6-8
Ammonia (mg L ⁻¹)	0-0.05	0-0.05	0-0.05
Nitrite (mg L ⁻¹)	0-0.01	0-0.01	0-0.01
Nitrate (mg L ⁻¹)	0-0.01	0-0.01	0-0.01
Temperature (°C)	29-33	28-32	28-32
Flow rate (L second ⁻¹)	11.7	11.7	11.7
Circulation rate (L second ⁻¹)	11.7	11.7	11.7

Data expressed in range. T200: 200 fish m⁻³, T250: 250 fish m⁻³, T300: 300 fish m⁻³.

Water quality is one of the important factors to be considered for intensified aquaculture systems, as sudden fluctuations may lead to mortality and losses for farmers [50-51]. During the experiment, water temperature, pH, total ammonia, and nitrite and nitrate concentrations were reported to be in the range of 28-33°C, 7.6-8.4, 0.00-0.001 mg L⁻¹, and 0.01 mg L⁻¹, respectively, which are suitable for the culture of Singhi catfish in intensified systems, as reported in previous studies [52].

3.5 Economics

The economic indices were analyzed and are presented in Table 4. The results indicated a significantly higher ($P \leq 0.05$) feed cost in the T300 group among operational costs, which contributed to a higher total investment. Other economic parameters, such as revenue, benefit-cost ratio, return on investment, and annual net profit, were higher in the T250 group, though these were statistically similar ($P > 0.05$) among other groups.

Table 4 Economics evaluation of a recirculating aquaculture system for *Heteropneustes fossilis* cultured in a commercial setup.

Sl. No.	Component	T200	T250	T300	P Value
A	Operational cost				
i.	Feed	7074.00±442.01 ^a	8910.00±309.32 ^b	10530.00±214.31 ^c	0.001
ii.	Seed	9000.00±0.00	11250.00±0.00	13500.00±0.00	
iii.	Manpower	2223.00±0.00	2223.00±0.00	2223.00±0.00	
iv.	Electricity	750.00±0.00	750.00±0.00	750.00±0.00	
B	Capital expenditure				
i.	Infrastructure, machinery, and equipment	31000.00±0.00	31000.00±0.00	31000.00±0.00	
ii.	Land	1000.00±0.00	1000.00±0.00	1000.00±0.00	
A+B	Total investment	51047.00±442.01 ^a	55133.00±309.32 ^b	59003.00±214.31 ^c	0.001
	Income	65228.47±1479.29	71752.71±1533.29	73131.53±3816.29	
	Revenue	14181.47±1400.47	16619.71±1733.14	14128.53±3775.05	
	Benefit-cost ratio	1.28±0.03	1.30±0.03	1.24±0.06	
	Return on investment	27.78±2.77	30.17±3.27	23.94±6.40	
	Annual net profit	70907.33±7002.35	83098.56±8665.72	70642.64±18875.25	

Data are expressed as mean ± standard error, $n = 3$ (one-way analysis of variance); mean values in the same row with different superscripts differ significantly ($P < 0.05$). T200: 200 fish m⁻³, T250: 250 fish m⁻³, T300: 300 fish m⁻³. Land area calculation: 1 decimal = 40.47 m².

Economic indices will help in understanding the economic potential of RAS-based culture system. In operational cost, feed and seed are major contributors [53], and in total investment, feed and seed costs contribute 13.3-18.4% and 17.7-23% of total investment, respectively. The experiment reported an increasing feed and seed costs with increased stocking density as per previous reports [17]. The highest revenue of Rs 16,619.71 ± 1,733.14 was obtained from the T250 group, though the values were non-significant among the treatments. The highest annual net profit of Rs 83,098.56 ± 8,665.70 was obtained from T250, which proves the system's economic efficiency and productivity. Previous studies have reported a net revenue of Rs. 14,040.00 from low-cost RAS-based Singhi nursery units [35]. The highest benefit-cost ratio of 1.30 was recorded in T250, and a similar result was also observed in previous studies [43].

3.6 Water productivity

Water productivity indices were calculated and are presented in Table 5. The result indicated a statistically similar ($P \geq 0.05$) TWU, CWU, TWP, GTWP, NTWP, and PWP in all treatments. TWUI was statistically different ($P \leq 0.05$) among treatments, with the lowest reported in T300.

Table 5 The water productivity and utilization of *Heteropneustes fossilis* cultured in the recirculating aquaculture system.

Parameters	T200	T250	T300	P value
TWU	42.99±0.00	42.99±0.00	42.99±0.00	-
CWU	14.99±0.00	14.99±0.00	14.99±0.00	-
Consumptive water use index	0.0029±0.01 ^c	0.0025±0.01 ^b	0.0020±0.01 ^a	0.001
Total water use index	0.0082±0.01 ^c	0.0068±0.01 ^b	0.0058±0.01 ^a	0.001
GTWP	1517.29±34.41	1669.06±35.66	1701.13±88.78	0.139
NTWP	329.88±32.58	386.59±40.32	328.65±87.81	0.740
PWP	2.53±0.06	2.78±0.06	2.84±0.15	0.139

Data are expressed as mean ± standard error, $n = 3$ (one-way analysis of variance); mean values in the same row with different superscripts differ significantly ($P < 0.05$); economic values are represented in USD (1\$ = 87.03 INR). T200: 200 fish m⁻³, T250: 250 fish m⁻³, T300: 300 fish m⁻³. TWU, Total Water Use (m³); CWU, Consumptive Water Use (m³); GTWP, Gross Total Water Productivity (total gross income in USD m⁻³ water volume); NTWP, Net Total Water Productivity (total net income in USD m⁻³ water volume); PWP, Physical Water Productivity (yield in kg m⁻³ water volume).

Water productivity indices were analyzed to understand the water-use efficiency of the aquaculture system [54]. TWU is an indicator of the total water required for the study, whereas CWU indicates the amount of water replenished to compensate for losses during exchange [55]. The results indicated similar TWU and CWU across all treatments; therefore, a single culture cycle of Singhi larvae reared in RAS requires around 42,990 L of water. The highest survivability was recorded in T250; consequently, the highest TWUI was also observed in the treatment, indicating maximum utilization of the water area. TWUI further indicated that the culture of a single Singhi fry in RAS requires around 8.2 L of water [27,56]. The highest GTWP and PWP were recorded in T300, indicating highest income and productivity from the system [24], but NTWP in T250 indicated better profit at 250 fish m⁻³ stocking density. PWP indicated that a maximum of 2.78 kg of fish can be harvested from 1,000 L of water at 250 fish m⁻³ stocking density. Therefore, it can be concluded that 250 fish m⁻³ is ideal in terms of sustainable and water-efficient larval rearing of Singhi in a RAS-based system.

4. Conclusions

The current research was focused on assessing the optimum stocking density of *H. fossilis*, which can promote sustainable and economically feasible aquaculture. The findings indicated that a stocking density of 250 fish m⁻³ significantly improved growth performance, survival, and overall economic viability of *H. fossilis* cultured in a RAS-based system.

5. Additional Information

5.1 Author Contributions

All authors have reviewed the final version to be published and agreed to be accountable for all aspects of the work.

Concept and design: Md Aklakur, Md Imran Shah, Udipta Roy, Ashutosh Dharmendra Deo.

Acquisition, analysis, or interpretation of data: Md Aklakur, Md Imran Shah, Udipta Roy, Ashutosh Dharmendra Deo.

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Critical review of the manuscript for important intellectual content: Md Aklakur, Md Imran Shah, Udipta Roy, Ashutosh Dharmendra Deo.

Supervision: Md Aklakur.

5.2 Disclosures

Plant subjects: All authors have confirmed that this study did not involve any herbarium specimen. Human subjects: All authors have confirmed that this study did not involve human participants or tissue. Animal subjects: The study involves the use of animal subjects. All procedures were conducted in accordance with the Prevention of Cruelty to Animals Act, 1960, approved by CPCSEA (Committee for the Purpose of Control and Supervision of Experiments on Animals) and conducted under the registration number approved to the institute (ICAR-CIFE) with respect to registration number 524/GO/Re/S/02/CPCSEA. The research was conducted following all rules and regulations recommended as per the certificate. The ARRIVE guidelines have been followed in reporting the study. ethics committee, protocol no registration number 524/GO/Re/S/02/CPCSEA. Conflicts of interest: In compliance with the ICMJE uniform disclosure form, all authors declare the following: Payment/services info: All authors have declared that no financial support was received from any organization for the submitted work. Financial relationships: All authors have declared that they have no financial relationships at present or within the previous three years with any organizations that might have an interest in the submitted work. Other relationships: All authors have declared that there are no other relationships or activities that could appear to have influenced the submitted work.

5.3 Data Availability Statements

The datasets generated and/or analyzed during the current study are not publicly available as the research is conducted under ICAR-CIFE, RRTC, Motipur, through a commercial biofloc fish farm, a government body that do not share the raw data to public forums, but it could be accessed from the corresponding author on reasonable and valid request.

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